$x = \frac{mx_2 + nx_1}{m+n}$

$\therefore\ 11 = \frac{9m+15n}{m+n}$

$\therefore\ 11m + 11n = 9m + 15n$

$\therefore\ 2m = 4n$

$\therefore\ \frac{m}{n} = \frac{4}{2} = \frac{2}{1}$

$\therefore$ The required ratio is $2 : 1$.

Similarly, find the ratio using y co-ordinates. Write the conclusion.

Ex. (4) Find the co-ordinates of the points of trisection of the segment joining the points A $(2,-2)$ and B$(-7,4)$.

(The two points that divide the line segment in three equal parts are called as points of trisection of the segment.)

Solution : Let points P and Q be the points of trisection of the line segment joining the points A and B.

Point P and Q divide line segment AB into three parts.

AP = PQ = QB ..... (I)

$\frac{AP}{PB} = \frac{AP}{PQ+QB} = \frac{AP}{AP+AP} = \frac{AP}{2AP} = \frac{1}{2}$ ..... From (I)

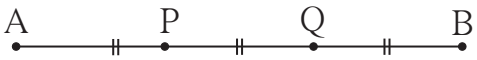


Fig. 5.18

Point P divides seg AB in the ratio 1:2.

x co-ordinate of point P $= \frac{1\times(-7)+2\times2}{1+2} = \frac{-7+4}{3} = \frac{-3}{3} = -1$

y co-ordinate of point P $= \frac{1\times4+2\times(-2)}{1+2} = \frac{4-4}{3} = \frac{0}{3} = 0$

Point Q divides seg AB in the ratio 2:1. $\therefore \frac{AQ}{QD} = \frac{2}{1}$

x co-ordinate of point Q $= \frac{2\times(-7)+1\times2}{2+1} = \frac{-14+2}{3} = \frac{-12}{3} = -4$

y co-ordinate of point Q $= \frac{2\times4+1\times-2}{2+1} = \frac{8-2}{3} = \frac{6}{3} = 2$

$\therefore$ co-ordinates of points of trisection are $(-1, 0)$ and $(-4, 2)$.

For more information :

See how the external division of the line segment joining points A and B takes place.

Let us see how the co-ordinates of point P can be found out if P divides the line segment joining points A(−4, 6) and B(5, 10) in the ratio 3:1 externally.

$\frac{AP}{PB} = \frac{3}{1}$ that is AP is larger than PB and A−B−P.

$\frac{AP}{PB} = \frac{3}{1}$ that is AP = $3k$, BP = k, then AB = $2k$

$\therefore \frac{AB}{BP} = \frac{2}{1}$

Now point B divides seg AP in the ratio 2 : 1.

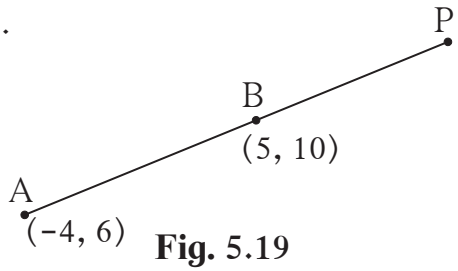


Fig. 5.19

We have learnt to find the coordinates of point P if co-ordinates of points A and B are known.

Practice set 5.2

1. Find the coordinates of point P if P divides the line segment joining the points A(−1,7) and B(4,−3) in the ratio 2 : 3.

2. In each of the following examples find the co-ordinates of point A which divides segment PQ in the ratio $a : b$.

(1) P(−3, 7), Q(1, −4), $a : b$ = 2 : 1

(2) P(−2, −5), Q(4, 3), $a : b$ = 3 : 4

(3) P(2, 6), Q(−4, 1), $a : b$ = 1 : 2

3. Find the ratio in which point T(−1, 6)divides the line segment joining the points P(−3, 10) and Q(6, −8).

4. Point P is the centre of the circle and AB is a diameter . Find the coordinates of point B if coordinates of point A and P are (2, −3) and (−2, 0) respectively.

5. Find the ratio in which point P(k, 7) divides the segment joining A(8, 9) and B(1, 2). Also find k .

6. Find the coordinates of midpoint of the segment joining the points (22, 20) and (0, 16).

7. Find the centroids of the triangles whose vertices are given below.

(1)(−7, 6), (2, −2), (8, 5)

(2) (3, −5), (4, 3), (11, −4)

(3) (4, 7), (8, 4), (7, 11)

8. In Δ ABC, G $(-4, -7)$ is the centroid. If A $(-14, -19)$ and B$(3, 5)$ then find the co-ordinates of C.

9. A$(h, -6)$, B$(2, 3)$ and C$(-6, k)$ are the co-ordinates of vertices of a triangle whose centroid is G $(1, 5)$. Find h and k.

10. Find the co-ordinates of the points of trisection of the line segment AB with A$(2, 7)$ and B$(-4, -8)$.

11. If A $(-14, -10)$, B$(6, -2)$ is given, find the coordinates of the points which divide segment AB into four equal parts.

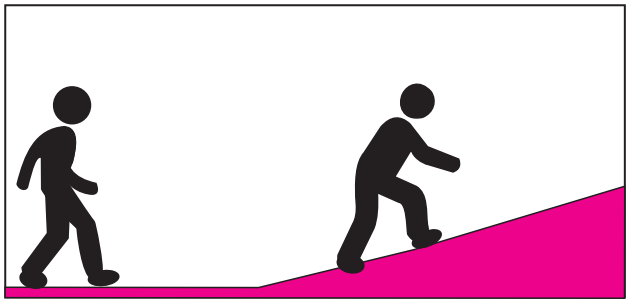
12. If A $(20, 10)$, B$(0, 20)$ are given, find the coordinates of the points which divide segment AB into five congruent parts.



Let's learn.

Slope of a line

When we walk on a plane road we need not exert much effort but while climbing up a slope we need more effort. In science, we have studied that while climbing up a slope we have to work against gravitational force.



In co-ordinate geometry, slope of a line is an important concept. We will learn it through the following activity.

Activity I :

In the figure points A$(-2, -5)$, B$(0, -2)$, C$(2, 1)$, D$(4, 4)$, E$(6, 7)$ lie on line l. Observe the table which is made with the help of coordinates of these points on line l.

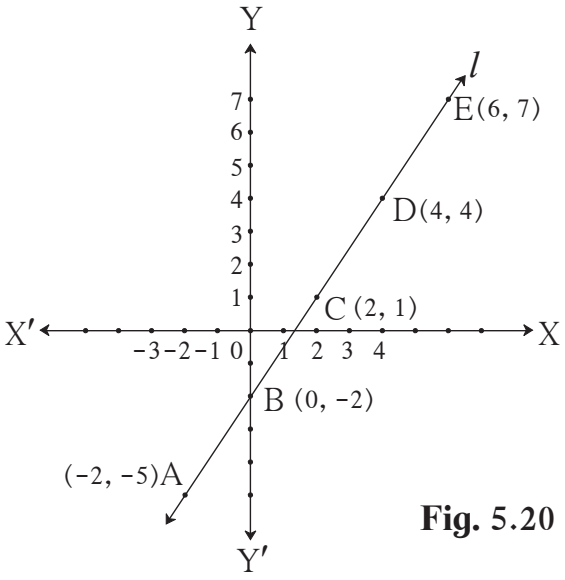


Fig. 5.20

Sr. No.	First point	Second point	Co-ordinates of first point (x_1, y_1)	Co-ordinates of second point (x_2, y_2)	$\frac{y_2 - y_1}{x_2 - x_1}$
1	C	E	(2, 1)	(6, 7)	$\frac{7-1}{6-2} = \frac{6}{4} = \frac{3}{2}$
2	A	D	(−2, −5)	(4, 4)	$\frac{4-(-5)}{4-(-2)} = \frac{9}{6} = \frac{3}{2}$
3	D	A	(4, 4)	(−2, −5)	$\frac{-5-4}{-2-4} = \frac{-9}{-6} = \frac{3}{2}$
4	B	C	--	--	--
5	C	A	--	--	--
6	A	C	--	--	--

Fill in the blank spaces in the above table. Similarly take some other pairs of points on line l and find the ratio $\frac{y_2 - y_1}{x_2 - x_1}$ for each pair.

From this activity, we understand that for any two points (x_1, y_1) and (x_2, y_2) on line l, the ratio $\frac{y_2 - y_1}{x_2 - x_1}$ is constant.

If (x_1, y_1) and (x_2, y_2) are any two points on line l, the ratio $\frac{y_2 - y_1}{x_2 - x_1}$ is called the slope of the line l.

Generally slope is shown by letter m.

$$\therefore \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

Activity II : In the figure, some points on line l, t and n are given. Find the slopes of those lines.

Now you will know,

(1) Slopes of line l and line t are positive.

(2) Slope of line n is negative.

(3) Slope of line t is more than slope of line l.

(4) Slopes of lines l and t which make acute angles with X- axis, are positive.

(5) Slope of line n making obtuse angle with X- axis is negative.

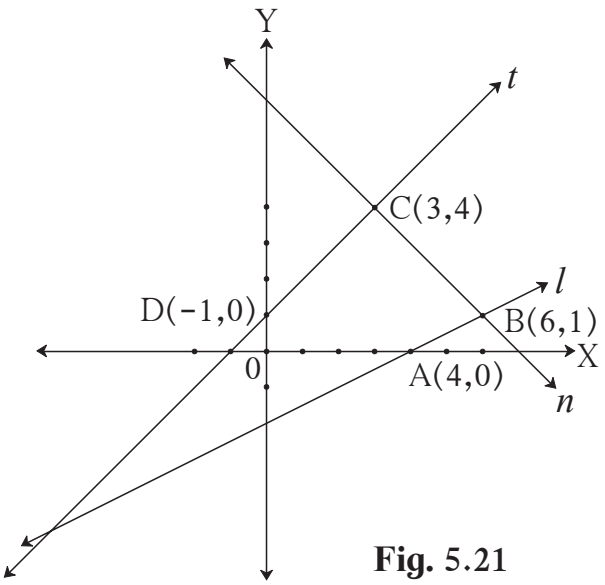


Fig. 5.21

Slopes of X-axis, Y-axis and lines parallel to axes.

In the figure 5.22, $(x_1, 0)$ and $(x_2, 0)$ are two points on the X- axis.

Slope of X- axis $= \frac{0-0}{x_2 - x_1} = 0$

In the same way $(0, y_1)$ and $(0, y_2)$ are two points on the Y- axis.

Slope of Y- axis $= \frac{y_2 - y_1}{0-0} = \frac{y_2 - y_1}{0}$,

But division by 0 is not possible.

$\therefore$ slope of Y- axis can not be determined.

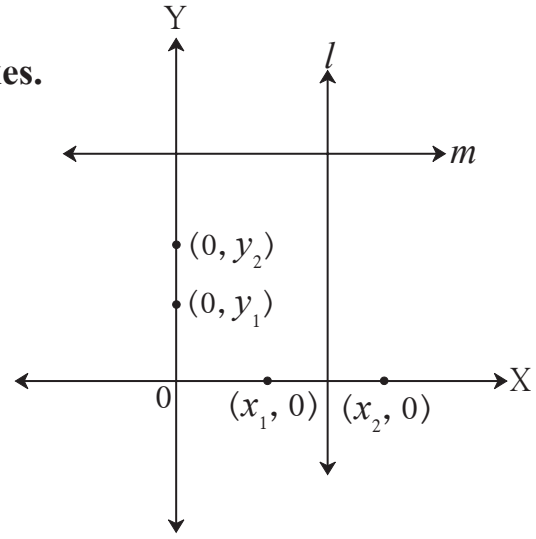


Fig. 5.22

Now try to find the slope of any line like line m which is parallel to X- axis. It will come out to be 0.

Similarly we cannot determine the slope of a line like line l which is parallel to Y- axis

Slope of line – using ratio in trigonometry

In the figure 5.23, points P(x_1, y_1) and Q (x_2, y_2) are on line l.

Line l intersects X axis in point T.

seg QS $\perp$ X- axis, seg PR $\perp$ seg QS $\therefore$ seg PR $\|$ seg TS ..... corresponding angle test

$\therefore$ QR $= y_2 - y_1$ and PR $= x_2 - x_1$

$\therefore \quad \frac{QR}{PR} = \frac{y_2 - y_1}{x_2 - x_1}$ ..... (I)

Line TQ makes an angle θ with the X- axis .

$\therefore \quad \frac{QR}{PR} = \tan\theta$ ..... (II)

$\therefore$ From (I) and (II), $\frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$

$\therefore \quad m = \tan\theta$

Now seg PR || seg TS, line l is transversal

$\therefore \quad \angle QPR = \angle QTS$ .....corresponding angles

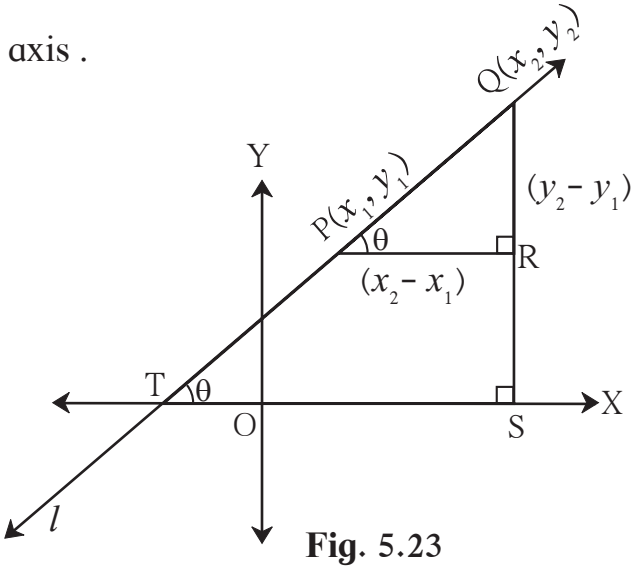


Fig. 5.23

From this, we can define slope in another way. The tan ratio of an angle made by the line with the positive direction of X-axis is called the slope of that line.

When any two lines have same slope, these lines make equal angles with the positive direction of X- axis.

$\therefore$ These two lines are parallel.

Slope of Parallel Lines

Activity :

In the figure 5.22 both line l and line t make angle θ with the positive direction of X- axis.

$\therefore$ line l || line t ..... corresponding angle test

Consider, point A(-3, 0) and point B(0, 3) on line l

Find the slope of line l.

Slope of line $l = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{\square - \square}{\square - \square} = \frac{\square}{\square}$

$= \square$

In the similar way, consider suitable points on the line t and find the slope of line t.

From this, you can verify that parallel lines have equal slopes.

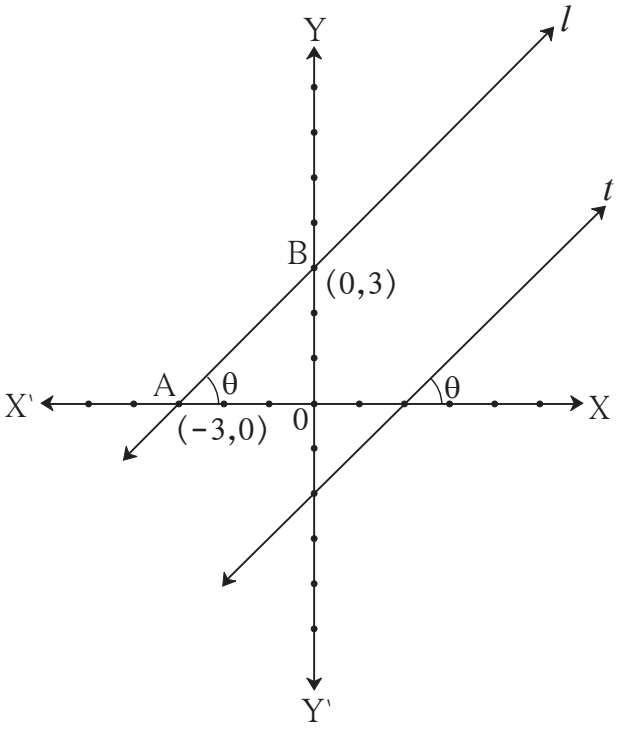


Fig. 5.24

Here $\theta = 45°$.

Use slope, $m = \tan\theta$ and verify that slopes of parallel lines are equal.

Similarly taking $\theta = 30°$, $\theta = 60°$ verify that slopes of parallel lines are equal.



Remember this!

The slope of X- axis and of any line parallel to X- axis is zero.

The slope of Y- axis and of any line parallel to Y- axis cannot be determined.

Solved Examples

EX. (1) Find the slope of the line passing through the points A $(-3, 5)$, and B $(4, -1)$

Solution : Let, $x_1 = -3$, $x_2 = 4$, $y_1 = 5$, $y_2 = -1$

$\therefore$ Slope of line AB $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-5}{4-(-3)} = \frac{-6}{7}$

EX. (2) Show that points P$(-2, 3)$, Q$(1, 2)$, R$(4, 1)$ are collinear.

Solution : P$(-2, 3)$, Q$(1, 2)$ and R$(4, 1)$ are given points

slope of line PQ $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-3}{1-(-2)} = -\frac{1}{3}$

Slope of line QR $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-2}{4-1} = -\frac{1}{3}$

Slope of line PQ and line QR is equal.

But point Q lies on both the lines.

$\therefore$ Point P, Q, R are collinear.

EX. (3) If slope of the line joining points P$(k, 0)$ and Q$(-3, -2)$ is $\frac{2}{7}$ then find k.

Solution : P$(k, 0)$ and Q$(-3, -2)$

Slope of line PQ $= \frac{-2-0}{-3-k} = \frac{-2}{-3-k}$

But slope of line PQ is given to be $\frac{2}{7}$.

$\therefore \frac{-2}{-3-k} = \frac{2}{7}$ $\therefore k = 4$

EX. (4) If A (6, 1), B (8, 2), C (9, 4) and D (7, 3) are the vertices of □ ABCD , show that □ ABCD is a parallelogram.

Solution : You know that Slope of line = $\frac{y_2 - y_1}{x_2 - x_1}$

Slope of line AB = $\frac{2-1}{8-6} = \frac{1}{2}$ ..... (I)

Slope of line BC = $\frac{4-2}{9-8} = 2$ ..... (II)

Slope of line CD = $\frac{3-4}{7-9} = \frac{1}{2}$ ..... (III)

Slope of line DA = $\frac{3-1}{7-6} = 2$ ..... (IV)

Slope of line AB = Slope of line CD ..... From (I) and (III)

∴ line AB || line CD

Slope of line BC = Slope of line DA ..... From (II) and (IV)

∴ line BC || line DA

Both the pairs of opposite sides of the quadrilateral are parallel

∴ □ ABCD is a parallelogram.

Practice set 5.3

1. Angles made by the line with the positive direction of X-axis are given. Find the slope of these lines.

(1) 45° (2) 60° (3) 90°

2. Find the slopes of the lines passing through the given points.

(1) A (2, 3) , B (4, 7) (2) P (−3, 1) , Q (5, −2)

(3) C (5, −2) , D (7, 3) (4) L (−2, −3) , M (−6, −8)

(5) E(−4, −2) , F (6, 3) (6) T (0, −3) , S (0, 4)

3. Determine whether the following points are collinear.

(1) A(−1, −1), B(0, 1), C(1, 3) (2) D(−2, −3), E(1, 0), F(2, 1)

(3) L(2, 5), M(3, 3), N(5, 1) (4) P(2, −5), Q(1, −3), R(−2, 3)

(5) R(1, −4), S(−2, 2), T(−3, 4) (6) A(−4, 4), K(−2, $\frac{5}{2}$), N(4, −2)

4. If A (1, −1),B (0, 4),C (−5, 3) are vertices of a triangle then find the slope of each side.

5. Show that A (−4, −7),B (−1, 2), C (8, 5) and D (5, −4) are the vertices of a parallelogram.

6. Find k, if R(1, −1), S (−2, k) and slope of line RS is −2.

7. Find k, if B(k, −5), C (1, 2) and slope of the line is 7.

8. Find k, if PQ || RS and P(2, 4), Q (3, 6), R(3, 1), S(5, k) .

Problem set 5

1. Fill in the blanks using correct alternatives.

(1) Seg AB is parallel to Y−axis and coordinates of point A are (1,3) then co-ordinates of point B can be …… .

(A) (3,1) (B) (5,3) (C) (3,0) (D) (1,−3)

(2) Out of the following, point …… lies to the right of the origin on X− axis.

(A) (−2,0) (B) (0,2) (C) (2,3) (D) (2,0)

(3) Distance of point (−3,4) from the origin is …… .

(A) 7 (B) 1 (C) 5 (D) −5

(4) A line makes an angle of 30° with the positive direction of X− axis. So the slope of the line is …… .

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

2. Determine whether the given points are collinear.

(1) A(0,2) , B(1,−0.5), C(2,−3)

(2) P(1, 2) , Q(2, $\frac{8}{5}$) , R(3, $\frac{6}{5}$)

(3) L(1,2) , M(5,3) , N(8,6)

3. Find the coordinates of the midpoint of the line segment joining P(0,6) and Q(12,20).

4. Find the ratio in which the line segment joining the points A(3,8) and B(−9, 3) is divided by the Y− axis.

5. Find the point on X−axis which is equidistant from P(2,−5) and Q(−2,9).

6. Find the distances between the following points.

(i) A(a, 0), B(0, a) (ii) P(−6, −3), Q(−1, 9) (iii) R(−3a, a), S(a, −2a)

7. Find the coordinates of the circumcentre of a triangle whose vertices are (−3,1), (0,−2) and (1,3)

8. In the following examples, can the segment joining the given points form a triangle ? If triangle is formed, state the type of the triangle considering sides of the triangle.

(1) L(6,4) , M(−5,−3) , N(−6,8)

(2) P(−2,−6) , Q(−4,−2), R(−5,0)

(3) A($\sqrt{2}$, $\sqrt{2}$), B($-\sqrt{2}$, $-\sqrt{2}$), C($-\sqrt{6}$, $\sqrt{6}$)

9. Find k if the line passing through points P(−12,−3) and Q(4, k)has slope $\frac{1}{2}$.

10. Show that the line joining the points A(4, 8) and B(5, 5) is parallel to the line joining the points C(2,4) and D(1,7).

11. Show that points P(1,−2), Q(5,2), R(3,−1), S(−1,−5) are the vertices of a parallelogram

12. Show that the □ PQRS formed by P(2,1), Q(−1,3), R(−5,−3) and S(−2,−5) is a rectangle

13. Find the lengths of the medians of a triangle whose vertices are A(−1, 1), B(5, −3) and C(3, 5) .

14⋆. Find the coordinates of centroid of the triangles if points D(−7, 6), E(8, 5) and F(2, −2) are the mid points of the sides of that triangle.

15. Show that A(4, −1), B(6, 0), C(7, −2) and D(5, −3)are vertices of a square.

16. Find the coordinates of circumcentre and radius of circumcircle of Δ ABC if A(7, 1), B(3, 5) and C(2, 0) are given.

17. Given A(4,−3), B(8,5). Find the coordinates of the point that divides segment AB in the ratio 3:1.

18⋆. Find the type of the quadrilateral if points A(−4, −2), B(−3, −7) C(3, −2) and D(2, 3) are joined serially.

19⋆. The line segment AB is divided into five congruent parts at P, Q, R and S such that A−P−Q−R−S−B. If point Q(12, 14) and S(4, 18) are given find the coordinates of A, P, R,B.

20. Find the coordinates of the centre of the circle passing through the points P(6,−6), Q(3,−7)and R(3,3).

21⋆. Find the possible pairs of coordinates of the fourth vertex D of the parallelogram, if three of its vertices are A(5,6), B(1,−2)and C(3,−2).

22. Find the slope of the diagonals of a quadrilateral with vertices A(1,7), B(6,3), C(0,−3) and D(−3,3).

❐❐❐

PUX2MU

6 Trigonometry

Let's study.

- Trigonometric ratios
- Angle of elevation and angle of depression
- Trigonometric identities
- Problems based on heights and distances

Let's recall.

1. Fill in the blanks with reference to figure 6.1 .

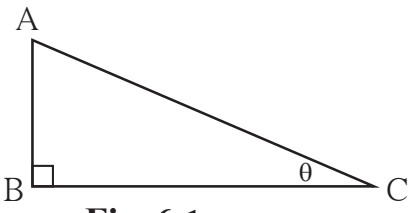


Fig. 6.1

$\sin\theta = \frac{\square}{\square}$, $\cos\theta = \frac{\square}{\square}$,

$\tan\theta = \frac{\square}{\square}$

2. Complete the relations in ratios given below .

(i) $\frac{\sin\theta}{\cos\theta} = \square$ (ii) $\sin\theta = \cos(90 - \square)$

(iii) $\cos\theta = \sin(90 - \square)$ (iv) $\tan\theta \times \tan(90 - \theta) = \square$

3. Complete the equation.

$\sin^2\theta + \cos^2\theta = \square$

4. Write the values of the following trigonometric ratios.

(i) $\sin 30° = \frac{1}{\square}$ (ii) $\cos 30° = \frac{\square}{\square}$ (iii) $\tan 30° = \frac{\square}{\square}$

(iv) $\sin 60° = \frac{\square}{\square}$ (v) $\cos 45° = \frac{\square}{\square}$ (vi) $\tan 45° = \square$

In std IX, we have studied some trigonometric ratios of some acute angles. Now we are going to study some more trigonometric ratios of acute angles.

Let's learn.

cosec, sec and cot ratios

Multiplicative inverse or the reciprocal of sine ratio is called cosecant ratio. It is written in brief as cosec. $\therefore \operatorname{cosec}\theta = \frac{1}{\sin\theta}$

Similarly, multiplicative inverses or reciprocals of cosine and tangent ratios are called "secant" and "cotangent" ratios respectively. They are written in brief as sec and cot.

$\therefore \sec\theta = \frac{1}{\cos\theta}$ and $\cot\theta = \frac{1}{\tan\theta}$

In figure 6.2,

$$\sin\theta = \frac{AB}{AC}$$

$$\therefore \operatorname{cosec}\theta = \frac{1}{\sin\theta}$$

$$= \frac{1}{\frac{AB}{AC}}$$

$$= \frac{AC}{AB}$$

It means,

$$\operatorname{cosec}\theta = \frac{\text{hypotenuse}}{\text{opposite side}}$$

$$\tan\theta = \frac{AB}{BC}$$

$$\therefore \cot\theta = \frac{1}{\tan\theta}$$

$$= \frac{1}{\frac{AB}{BC}}$$

$$\cot\theta = \frac{BC}{AB} = \frac{\text{adjacent side}}{\text{opposite side}}$$

$$\cos\theta = \frac{BC}{AC}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$= \frac{1}{\frac{BC}{AC}}$$

$$= \frac{AC}{BC}$$

It means,

$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent side}}$$

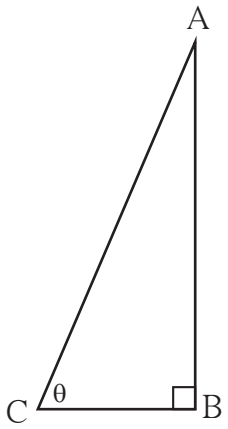


Fig. 6.2

You know that,

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\therefore \cot\theta = \frac{1}{\tan\theta}$$

$$= \frac{1}{\frac{\sin\theta}{\cos\theta}}$$

$$= \frac{\cos\theta}{\sin\theta}$$

$$\therefore \cot\theta = \frac{\cos\theta}{\sin\theta}$$

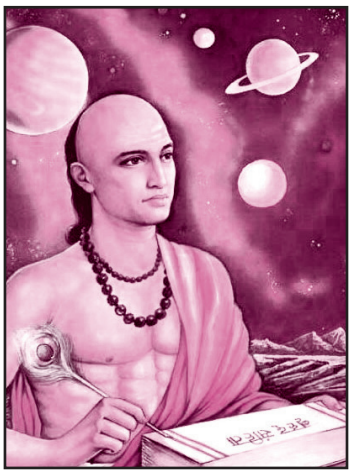


Remember this!

The relation between the trigonometric ratios,
according to the definitions of cosec, sec and cot ratios

- $\frac{1}{\sin\theta} = \text{cosec}\ \theta$ $\quad\therefore\ \sin\theta \times \text{cosec}\ \theta = 1$
- $\frac{1}{\cos\theta} = \sec\theta$ $\quad\therefore\ \cos\theta \times \sec\theta = 1$
- $\frac{1}{\tan\theta} = \cot\theta$ $\quad\therefore\ \tan\theta \times \cot\theta = 1$

For more information :



The great Indian mathematician Aryabhata was born in 476 A.D. in Kusumpur which was near Patna in Bihar. He has done important work in Arithmetic, Algebra and Geometry. In the book 'Aryabhatiya' he has written many mathematical formulae. For example,

(1) In an Arithmetic Progression, formulae for n^{th} term and the sum of first n terms.

(2) The formula to approximate $\sqrt{2}$

(3) The correct value of π upto four decimals, $\pi = 3.1416$.

In the study of Astronomy he used trigonometry and the sine ratio of an angle for the first time.

Comparing with the mathematics in the rest of the world at that time, his work was great and was studied all over India and was carried to Europe through Middle East.

Most observers at that time believed that the earth is immovable and the Sun, the Moon and stars move arround the earth. But Aryabhata noted that when we travel in a boat on the river, objects like trees, houses on the bank appear to move in the opposite direction. 'Similarly', he said 'the Sun, Moon and the stars are observed by people on the earth to be moving in the opposite direction while in reality the Earth moves !'

On 19 April 1975, India sent the first satellite in the space and it was named 'Aryabhata' to commemorate the great Mathematician of India.

✻ The table of the values of trigonometric ratios of angles $0°, 30°, 45°, 60°$ and $90°$.

Trigonometric ratio	Angle (θ)				
	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\text{cosec}\,\theta = \frac{1}{\sin\theta}$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta = \frac{1}{\cos\theta}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot\theta = \frac{1}{\tan\theta}$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0



Let's learn.

Trigonometric identities

In the figure 6.3, Δ ABC is a right angled triangle, $\angle B = 90°$

(i) $\sin\theta = \frac{BC}{AC}$ (ii) $\cos\theta = \frac{AB}{AC}$

(iii) $\tan\theta = \frac{BC}{AB}$ (iv) $\text{cosec}\,\theta = \frac{AC}{BC}$

(v) $\sec\theta = \frac{AC}{AB}$ (vi) $\cot\theta = \frac{AB}{BC}$

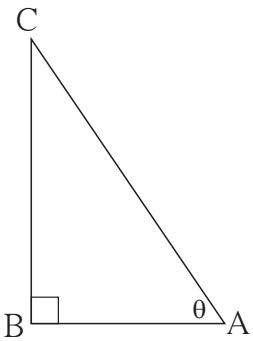


Fig. 6.3

By Pythagoras therom,

$BC^2 + AB^2 = AC^2$ . . . . .(I)

Dividing both the sides of (1) by AC^2

$$\frac{BC^2 + AB^2}{AC^2} = \frac{AC^2}{AC^2}$$

$\therefore \quad \frac{BC^2}{AC^2} + \frac{AB^2}{AC^2} = 1$

$\therefore \quad \left(\frac{BC}{AC}\right)^2 + \left(\frac{AB}{AC}\right)^2 = 1$

$\therefore (\sin\theta)^2 + (\cos\theta)^2 = 1$ .... [$(\sin\theta)^2$ is written as $\sin^2\theta$ and $(\cos\theta)^2$ is written as $\cos^2\theta$.]

$$\mathbf{\sin^2\theta + \cos^2\theta = 1} \quad \ldots\ldots\ldots \text{(II)}$$

Now dividing both the sides of equation (II) by $\sin^2\theta$

$$\frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

$$\mathbf{1 + \cot^2\theta = \operatorname{cosec}^2\theta} \quad \ldots\ldots\ldots \text{(III)}$$

Dividing both the sides of equation (II) by $\cos^2\theta$

$$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\mathbf{1 + \tan^2\theta = \sec^2\theta} \quad \ldots\ldots\ldots \text{(IV)}$$

Relations (II),(III), and (IV) are fundamental trigonometric identities.

Solved Examples

Ex. (1) If $\sin\theta = \frac{20}{29}$ then find $\cos\theta$

Solution : Method I

We have

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{20}{29}\right)^2 + \cos^2\theta = 1$

$\frac{400}{841} + \cos^2\theta = 1$

$\cos^2\theta = 1 - \frac{400}{841}$

$= \frac{441}{841}$

Taking square root of both sides.

$\cos\theta = \frac{21}{29}$

Method II

$\sin\theta = \frac{20}{29}$

from figure, $\sin\theta = \frac{AB}{AC}$

$\therefore$ AB = $20k$ and AC = $29k$

Let BC = x.

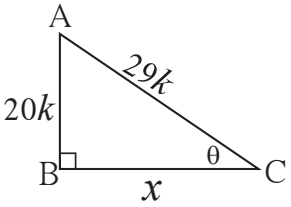


Fig. 6.4

According to Pythagoras therom,

$AB^2 + BC^2 = AC^2$

$(20k)^2 + x^2 = (29k)^2$

$400k^2 + x^2 = 841k^2$

$x^2 = 841k^2 - 400k^2$

$= 441k^2$

$\therefore \quad x = 21k$

$\therefore \cos\theta = \frac{BC}{AC} = \frac{21k}{29k} = \frac{21}{29}$

Ex. (2) If $\sec\theta = \frac{25}{7}$, find the value of $\tan\theta$.

Solution : Method I

we have ,

$1 + \tan^2\theta = \sec^2\theta$

$\therefore \quad 1 + \tan^2\theta = \left(\frac{25}{7}\right)^2$

$\therefore \quad \tan^2\theta = \frac{625}{49} - 1$

$= \frac{625 - 49}{49}$

$= \frac{576}{49}$

$\therefore \quad \tan\theta = \frac{24}{7}$

Method II

from the figure,

$\sec\theta = \frac{PR}{PQ}$

$\therefore \quad PQ = 7k,\ PR = 25k$

According to Pythagoras therom,

$PQ^2 + QR^2 = PR^2$

$\therefore \quad (7k)^2 + QR^2 = (25k)^2$

$\therefore \quad QR^2 = 625k^2 - 49k^2 = 576k^2$

$\therefore \quad QR = 24k$

Now, $\tan\theta = \frac{QR}{PQ} = \frac{24k}{7k} = \frac{24}{7}$

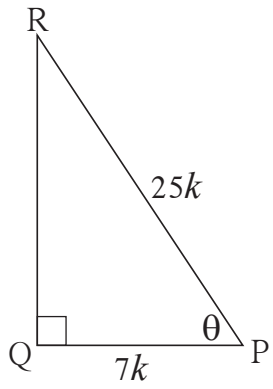


Fig. 6.5

Ex. (3) If $5\sin\theta - 12\cos\theta = 0$, find the values of $\sec\theta$ and $\operatorname{cosec}\theta$.

Solution : $5\sin\theta - 12\cos\theta = 0$

$\therefore 5\sin\theta = 12\cos\theta$

$\therefore \frac{\sin\theta}{\cos\theta} = \frac{12}{5}$

$\therefore \tan\theta = \frac{12}{5}$

we have,

$1 + \tan^2\theta = \sec^2\theta$

$\therefore 1 + \left(\frac{12}{5}\right)^2 = \sec^2\theta$

$\therefore 1 + \frac{144}{25} = \sec^2\theta$

$\therefore \frac{25 + 144}{25} = \sec^2\theta$

$\therefore \quad \sec^2\theta = \frac{169}{25}$

$\therefore \quad \sec\theta = \frac{13}{5}$

$\therefore \cos\theta = \frac{5}{13}$

Now, $\sin^2\theta + \cos^2\theta = 1$

$\therefore \sin^2\theta = 1 - \cos^2\theta$

$\therefore \sin^2\theta = 1 - \left(\frac{5}{13}\right)^2$

$= 1 - \frac{25}{169}$

$= \frac{144}{169}$

$\therefore \sin\theta = \frac{12}{13}$

$\therefore \operatorname{cosec}\theta = \frac{13}{12}$

Ex. (4) $\cos\theta = \frac{\sqrt{3}}{2}$ then find the value of $\frac{1-\sec\theta}{1+\operatorname{cosec}\theta}$.

Solution : **Method I**

$\cos\theta = \frac{\sqrt{3}}{2} \quad \therefore \sec\theta = \frac{2}{\sqrt{3}}$

$\sin^2\theta + \cos^2\theta = 1$

$\therefore \sin^2\theta + \left(\frac{\sqrt{3}}{2}\right)^2 = 1$

$\therefore \sin^2\theta = 1 - \frac{3}{4} = \frac{1}{4}$

$\therefore \sin\theta = \frac{1}{2} \quad \therefore \operatorname{cosec}\theta = 2$

$$\therefore \frac{1-\sec\theta}{1+\operatorname{cosec}\theta} = \frac{1-\frac{2}{\sqrt{3}}}{1+2}$$

$$= \frac{\frac{\sqrt{3}-2}{\sqrt{3}}}{3}$$

$$= \frac{\sqrt{3}-2}{3\sqrt{3}}$$

Method II

$\cos\theta = \frac{\sqrt{3}}{2}$

we know that, $\cos 30° = \frac{\sqrt{3}}{2}$.

$\therefore \quad \theta = 30°$

$\therefore \sec\theta = \sec 30° = \frac{2}{\sqrt{3}}$

$\operatorname{cosec}\theta = \operatorname{cosec} 30° = 2$

$$\therefore \frac{1-\sec\theta}{1+\operatorname{cosec}\theta} = \frac{1-\frac{2}{\sqrt{3}}}{1+2}$$

$$= \frac{\frac{\sqrt{3}-2}{\sqrt{3}}}{3}$$

$$= \frac{\sqrt{3}-2}{3\sqrt{3}}$$

Ex. (5) Show that $\sec x + \tan x = \sqrt{\frac{1+\sin x}{1-\sin x}}$

Solution : $\sec x + \tan x = \frac{1}{\cos x} + \frac{\sin x}{\cos x}$

$$= \frac{1+\sin x}{\cos x}$$

$$= \sqrt{\frac{(1+\sin x)^2}{\cos^2 x}}$$

$$= \sqrt{\frac{(1+\sin x)(1+\sin x)}{1-\sin^2 x}}$$

$$= \sqrt{\frac{(1+\sin x)(1+\sin x)}{(1-\sin x)(1+\sin x)}}$$

$$= \sqrt{\frac{1+\sin x}{1-\sin x}}$$

Ex. (6) Eliminate θ from given equations.

$x = a \cot\theta - b \operatorname{cosec}\theta$

$y = a \cot\theta + b \operatorname{cosec}\theta$

Solution : $x = a \cot\theta - b \operatorname{cosec}\theta$ ..... (I)

$y = a \cot\theta + b \operatorname{cosec}\theta$ ..... (II)

Adding equations (I) and (II).

$x + y = 2a \cot\theta$

$\therefore \cot\theta = \dfrac{x+y}{2a}$ ..... (III)

Subtracting equation (II) from (I) ,

$y - x = 2b \operatorname{cosec}\theta$

$\therefore \operatorname{cosec}\theta = \dfrac{y-x}{2b}$ ..... (IV)

Now, $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\therefore \left(\dfrac{y-x}{2b}\right)^2 - \left(\dfrac{y+x}{2a}\right)^2 = 1$

$\therefore \dfrac{(y-x)^2}{4b^2} - \dfrac{(y+x)^2}{4a^2} = 1$

or $\left(\dfrac{y-x}{b}\right)^2 - \left(\dfrac{y+x}{a}\right)^2 = 4$

Practice set 6.1

1. If $\sin\theta = \dfrac{7}{25}$, find the values of $\cos\theta$ and $\tan\theta$.

2. If $\tan\theta = \dfrac{3}{4}$, find the values of $\sec\theta$ and $\cos\theta$.

3. If $\cot\theta = \dfrac{40}{9}$, find the values of $\operatorname{cosec}\theta$ and $\sin\theta$.

4. If $5\sec\theta - 12\operatorname{cosec}\theta = 0$, find the values of $\sec\theta$, $\cos\theta$ and $\sin\theta$.

5. If $\tan\theta = 1$ then, find the values of $\dfrac{\sin\theta + \cos\theta}{\sec\theta + \operatorname{cosec}\theta}$.

6. Prove that:

(1) $\dfrac{\sin^2\theta}{\cos\theta} + \cos\theta = \sec\theta$

(2) $\cos^2\theta(1 + \tan^2\theta) = 1$

(3) $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = \sec\theta - \tan\theta$

(4) $(\sec\theta - \cos\theta)(\cot\theta + \tan\theta) = \tan\theta \sec\theta$

(5) $\cot\theta + \tan\theta = \operatorname{cosec}\theta \sec\theta$

(6) $\frac{1}{\sec\theta - \tan\theta} = \sec\theta + \tan\theta$

(7) $\sin^4\theta - \cos^4\theta = 1 - 2\cos^2\theta$

(8) $\sec\theta + \tan\theta = \frac{\cos\theta}{1-\sin\theta}$

(9) If $\tan\theta + \frac{1}{\tan\theta} = 2$, then show that $\tan^2\theta + \frac{1}{\tan^2\theta} = 2$

(10) $\frac{\tan A}{(1+\tan^2 A)^2} + \frac{\cot A}{(1+\cot^2 A)^2} = \sin A \cos A$

(11) $\sec^4 A (1 - \sin^4 A) - 2\tan^2 A = 1$

(12) $\frac{\tan\theta}{\sec\theta - 1} = \frac{\tan\theta + \sec\theta + 1}{\tan\theta + \sec\theta - 1}$



Let's learn.

Application of trigonometry

Many times we need to know the height of a tower, building, tree or distance of a ship from the lighthouse or width of a river etc.

We cannot measure them actually but we can find them with the help of trigonometric ratios.

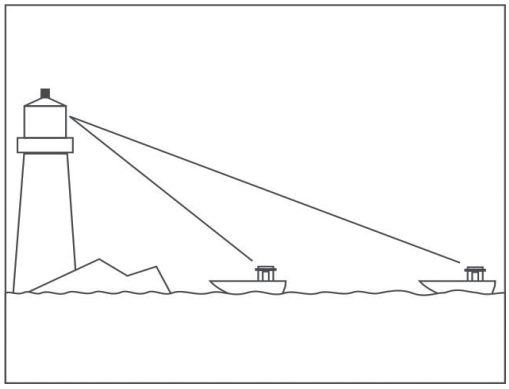


Fig. 6.6

For the purpose of computation, we draw a rough sketch to show the given information. 'Trees', 'hills' or 'towers' are vertical objects, so we shall represent them in the figure by segments which are perpendicular to the ground. We will not consider height of the observer and we shall normally regard observer's line of vision to be parallel to the horizontal level.

Let us study a few related terms.

(i) Line of vision : If the observer is standing at the location 'A', looking at an object 'B' then the line AB is called line of the vision.

(ii) Angle of elevation :

If an observer at A, observes the point B which is at a level higher than A and AM is the horizontal line, then ∠ BAM is called the angle of elevation.

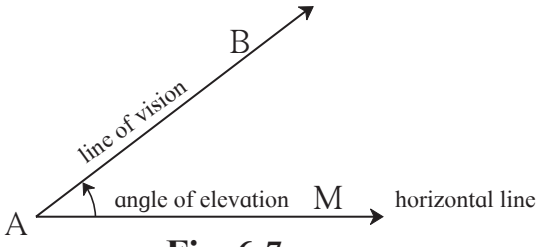


Fig. 6.7

(iii) Angle of depression :

If an observer at A, observes the point C which is at a level lower than A and AM is the horizontal line, the ∠ MAC is called the angle of depression.

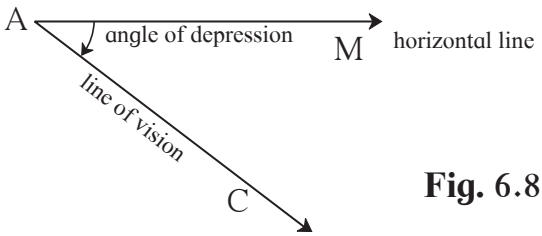


Fig. 6.8

When we see above the horizontal line, the angle formed is the angle of elevation. When we see below the horizontal line, the angle formed is the angle of depression.

Solved Examples

Ex. (1) An observer at a distance of 10 m from a tree looks at the top of the tree, the angle of elevation is 60°. What is the height of the tree ? ($\sqrt{3}$ = 1.73)

Solution : In figure 6.9, AB = h = height of the tree.

BC = 10 m, distance of the observer from the tree .

Angle of elevation (θ) = ∠ BCA = 60°

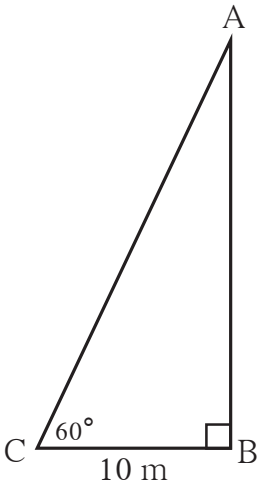


Fig. 6.9

from figure, $\tan\theta = \frac{AB}{BC}$ ..... (I)

$\tan 60° = \sqrt{3}$ ..... (II)

$\therefore \frac{AB}{BC} = \sqrt{3}$ ..... from equation (I)and (II)

$\therefore$ AB = BC $\sqrt{3}$ = 10 $\sqrt{3}$

$\therefore$ AB = 10 × 1.73 = 17.3 m

$\therefore$ height of the tree is 17.3m.

Ex. (2) From the top of a building, an observer is looking at a scooter parked at some distance away, makes an angle of depression of 30°. If the height of the building is 40m, find how far the scooter is from the building. ($\sqrt{3}$ = 1.73)

Solution: In the figure 6.10, AB is the building. A scooter is at C which is 'x' m away from the building.

In figure, 'A' is the position of the observer.

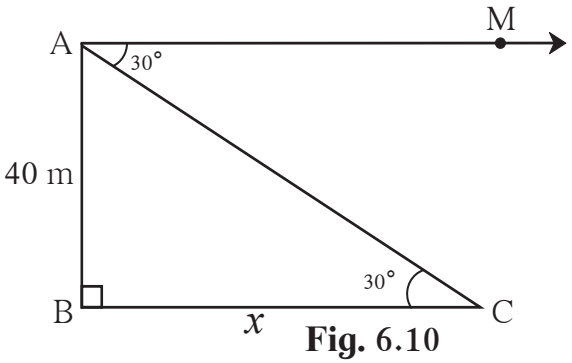


Fig. 6.10

AM is the horizontal line and ∠ MAC is the angle of depression.

∠ MAC and ∠ ACB are alternate angles.

from fig, $\tan 30° = \frac{AB}{BC}$

$\therefore \quad \frac{1}{\sqrt{3}} = \frac{40}{x}$

$\therefore \quad x = 40\sqrt{3}$

$= 40 \times 1.73$

$= 69.20$ m.

$\therefore$ the scooter is 69.20 m. away from the building.

Ex. (3) To find the width of the river, a man observes the top of a tower on the opposite bank making an angle of elevation of 61°. When he moves 50m backword from bank and observes the same top of the tower, his line of vision makes an angle of elevation of 35°. Find the height of the tower and width of the river. (tan61° = 1.8, tan35° = 0.7)

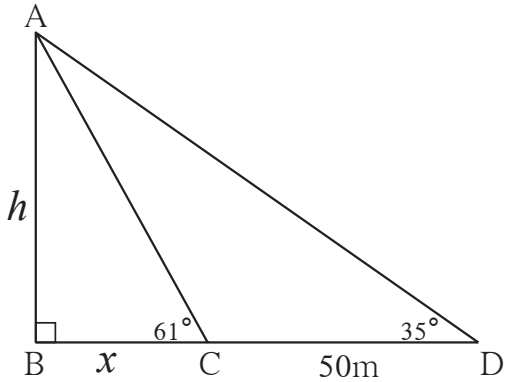


Fig. 6.11

Solution : seg AB shows the tower on the opposite bank. 'A' is the top of the tower and seg BC shows the width of the river. Let 'h' be the height of the tower and 'x' be the width of the river.

from figure, $\tan 61° = \frac{h}{x}$

$\therefore\ 1.8 = \frac{h}{x}$

$h = 1.8 \times x$

$10h = 18x$ ..... (I)..... multipling by 10

In right angled Δ ABD,

$\tan 35 = \frac{h}{x + 50}$

$0.7 = \frac{h}{x + 50}$

$\therefore\ \ h = 0.7\ (x + 50)$

$\therefore\ \ 10h = 7\ (x + 50)$ ..... (II)

$\therefore$ from equations (I) and (II) ,

$18x = 7(x + 50)$

$\therefore\ 18x = 7x + 350$

$\therefore\ 11x = 350$

$\therefore\ \ x = \frac{350}{11} = 31.82$

Now, $h = 1.8x = 1.8 \times 31.82$

$= 57.28$ m.

$\therefore$ width of the river = 31.82 m and height of tower = 57.28 m

Ex. (4) Roshani saw an eagle on the top of a tree at an angle of elevation of 61°, while she was standing at the door of her house. She went on the terrace of the house so that she could see it clearly. The terrace was at a height of 4m. While observing the eagle from there the angle of elevation was 52°. At what height from the ground was the eagle ?
(Find the answer correct upto nearest integer)

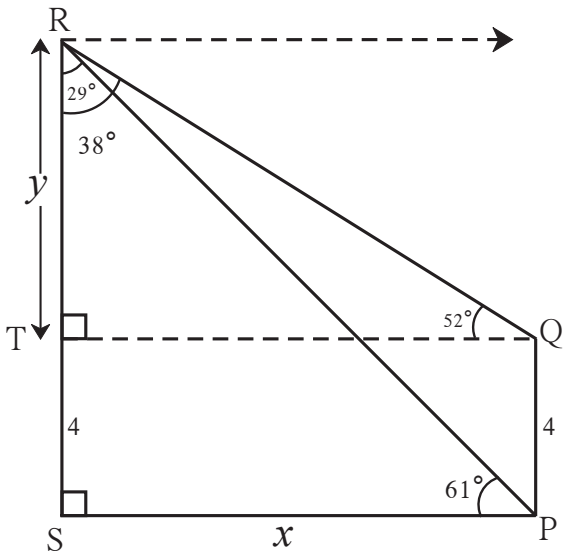


Fig. 6.12

(tan 61° = 1.80, tan 52° = 1.28, tan 29° = 0.55, tan 38° = 0.78)

Solution : In figure 6.12, PQ is the house and SR is the tree. The eagle is at R.

Draw seg QT $\perp$ seg RS.

$\therefore$ $\square$ TSPQ is a rectangle.

Let SP = x and TR = y

Now in Δ RSP, $\angle$ PRS = $90° - 61° = 29°$

and in Δ RTQ, $\angle$ QRT = $90° - 52° = 38°$

$\therefore$ $\tan \angle \text{PRS} = \tan 29° = \frac{\text{SP}}{\text{RS}}$

$\therefore$ $0.55 = \frac{x}{y+4}$

$\therefore$ $x = 0.55(y + 4)$ ..... (I)

Similarly, $\tan \angle \text{QRT} = \frac{\text{TQ}}{\text{RT}}$

$\therefore$ $\tan 38° = \frac{x}{y}$ ..... [$\because$ SP = TQ = x]

$\therefore$ $0.78 = \frac{x}{y}$

$\therefore$ $x = 0.78y$ ..... (II)

$\therefore$ $0.78y = 0.55(y + 4)$ ..... from (I) and (II)

$\therefore$ $78y = 55(y + 4)$

$\therefore$ $78y = 55y + 220$

$\therefore$ $23y = 220$

$\therefore$ $y = 9.565 = 10$ (upto nearest integer)

$\therefore$ RS = $y + 4 = 10 + 4 = 14$

$\therefore$ the eagle was at a height of 14 metre from the ground.

Ex. (5) A tree was broken due to storm. Its broken upper part was so inclined that its top touched the ground making an angle of $30°$ with the ground.The distance from the foot of the tree and the point where the top touched the ground was 10 metre. What was the height of the tree.

Solution: As shown in figure 6.13, suppose AB is the tree. It was broken at 'C' and its top touched at 'D'.

$\angle$ CDB = 30°, BD = 10 m, BC = x m

CA= CD = y m

In right angled Δ CDB,

$$\tan 30° = \frac{BC}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{x}{10}$$

$$x = \frac{10}{\sqrt{3}}$$

$$y = \frac{20}{\sqrt{3}}$$

$$x + y = \frac{10}{\sqrt{3}} + \frac{20}{\sqrt{3}}$$

$$= \frac{30}{\sqrt{3}}$$

$$x + y = 10\sqrt{3}$$

$\therefore$ height of the tree was $10\sqrt{3}$ m.

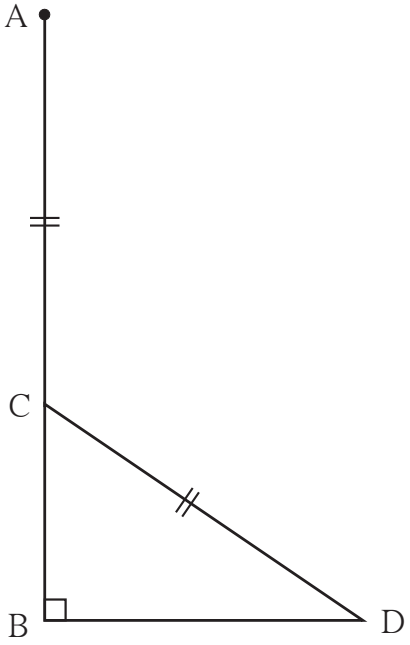


Fig. 6.13

Practice set 6.2

1. A person is standing at a distance of 80m from a church looking at its top. The angle of elevation is of 45°. Find the height of the church.

2. From the top of a lighthouse, an observer looking at a ship makes angle of depression of 60°. If the height of the lighthouse is 90 metre, then find how far the ship is from the lighthouse. ($\sqrt{3}$ =1.73)

3. Two buildings are facing each other on a road of width 12 metre. From the top of the first building, which is 10 metre high, the angle of elevation of the top of the second is found to be 60°. What is the height of the second building?

4. Two poles of heights 18 metre and 7 metre are erected on a ground. The length of the wire fastened at their tops in 22 metre. Find the angle made by the wire with the horizontal.

5. A storm broke a tree and the treetop rested 20 m from the base of the tree, making an angle of 60° with the horizontal. Find the height of the tree.

6. A kite is flying at a height of 60 m above the ground. The string attached to the kite is tied at the ground. It makes an angle of 60° with the ground. Assuming that the string is straight, find the length of the string. ($\sqrt{3}$ =1.73)

Problem set 6

1. Choose the correct alternative answer for the following questions.

(1) $\sin\theta \operatorname{cosec}\theta$ = ?

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

(2) $\operatorname{cosec}45^\circ$ = ?

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{2}{\sqrt{3}}$

(3) $1 + \tan^2\theta$ = ?

(A) $\cot^2\theta$ (B) $\operatorname{cosec}^2\theta$ (C) $\sec^2\theta$ (D) $\tan^2\theta$

(4) When we see at a higher level , from the horizontal line,angle formed is……. .

(A) angle of elevation. (B) angle of depression.

(C) 0 (D) straight angle.

2. If $\sin\theta = \frac{11}{61}$, find the values of $\cos\theta$ using trigonometric identity.

3. If $\tan\theta = 2$, find the values of other trigonometric ratios.

4. If $\sec\theta = \frac{13}{12}$, find the values of other trigonometric ratios.

5. Prove the following.

(1) $\sec\theta\,(1 - \sin\theta)\,(\sec\theta + \tan\theta) = 1$

(2) $(\sec\theta + \tan\theta)\,(1 - \sin\theta) = \cos\theta$

(3) $\sec^2\theta + \operatorname{cosec}^2\theta = \sec^2\theta \times \operatorname{cosec}^2\theta$

(4) $\cot^2\theta - \tan^2\theta = \operatorname{cosec}^2\theta - \sec^2\theta$

(5) $\tan^4\theta + \tan^2\theta = \sec^4\theta - \sec^2\theta$

(6) $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 2\sec^2\theta$

(7) $\sec^6 x - \tan^6 x = 1 + 3\sec^2 x \times \tan^2 x$

(8) $\frac{\tan\theta}{\sec\theta+1} = \frac{\sec\theta-1}{\tan\theta}$

(9) $\frac{\tan^3\theta-1}{\tan\theta-1} = \sec^2\theta + \tan\theta$

(10) $\frac{\sin\theta-\cos\theta+1}{\sin\theta+\cos\theta-1}=\frac{1}{\sec\theta-\tan\theta}$

6. A boy standing at a distance of 48 meters from a building observes the top of the building and makes an angle of elevation of 30°. Find the height of the building.

7. From the top of the light house, an observer looks at a ship and finds the angle of depression to be 30°.If the height of the light-house is 100 meters, then find how far the ship is from the light-house.

8. Two buildings are in front of each other on a road of width 15 meters. From the top of the first building, having a height of 12 meter, the angle of elevation of the top of the second building is 30°.What is the height of the second building?

9. A ladder on the platform of a fire brigade van can be elevated at an angle of 70° to the maximum. The length of the ladder can be extended upto 20m. If the platform is 2m above the ground, find the maximum height from the ground upto which the ladder can reach. (sin 70° = 0.94)

10 *. While landing at an airport, a pilot made an angle of depression of 20°. Average speed of the plane was 200 km/hr. The plane reached the ground after 54 seconds. Find the height at which the plane was when it started landing. (sin 20° = 0.342)

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